

Beginner's Guide to Basic Computer Technology and Command Line Usage

Introduction

Introduction to the basics of using a computer, understanding operating systems, and working with simple command-line tools. No prior experience is required.

1. Getting Started with Computers

1.1 What is a computer?

A computer is an electronic machine that processes data and performs tasks according to instructions given by the user.

1.2 Main Components of a Computer

- **Monitor:** Displays information.
- **System Unit (Case):** Contains internal components like CPU, RAM, storage.
- **Keyboard:** Used to enter text.
- **Mouse:** Used to navigate.

1.3 Turning the Computer on and Off

- Press the power button.
- Always shut down properly using the *Start Menu* → *Shut down*.

2. Operating System Basics

2.1 What is an Operating System (OS)?

Software that controls the hardware and allows you to work with applications. Examples: Windows, Linux, macOS.

2.2 Desktop Environment in Windows

- **Taskbar:** Shows running applications.
- **Start Menu:** Access programs and settings.
- **File Explorer:** Manage files and folders.

2.3 Creating and Managing Files

- Right-click → *New* → choose file type.
 - Use **Copy (Ctrl+C)**, **Paste (Ctrl+V)**, **Cut (Ctrl+X)**, **Rename** from right-click menu.
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3. Introduction to Command Line (CMD)

The Command Prompt allows you to type commands to interact with the system.

3.1 Opening Command Prompt

- Press **Windows + R**, type cmd, press **Enter**.

3.2 Basic Structure of a Command

command [options] [path or filename]

4. General CMD Commands (Step-by-Step)

4.1 Change Drive

C:> D:

Switches from C drive to D drive.

4.2 Clear Screen

cls

Clears the command window.

4.3 Show Folder Contents

dir

Lists all files and folders.

4.4 Show Contents Page by Page

dir /p

4.5 Show Wide Listing

dir /w

4.6 Check Date & Time

date
time

4.7 Check Version

ver

4.8 Get Help

help
help commandname

4.9 Exit CMD

exit

5. Directory (Folder) Commands

5.1 Create a Folder

md foldername

5.2 Remove a Folder

rd foldername

Folder must be **empty**.

5.3 Change Directory

cd folder-name

cd ...

cd \

6. File Commands

6.1 Create a File (Method 1)

echo Hello > file.txt

6.2 Create a File (Method 2)

copy con file.txt

(Type text → Press **Ctrl + C** to save)

6.3 Display File Content

type file.txt

6.4 Rename a File

ren oldname.txt newname.txt

6.5 Copy a File

copy file1.txt D:\folder

6.6 Copy Folder with Subfolders

xcopy source destination /s

7. Practical Exercises

Exercise 1

Create a folder named **StudentFiles** on the desktop.

Exercise 2

Inside it, create a text file named **info.txt**.

Exercise 3

Write your name in the file using CMD or GUI.

Exercise 4

Create a second folder called **Backup** and copy your file into it.

Exercise 5

Create Folders and Files with Arabic Language

8. Tips for Beginners

- Always double-check your spelling in CMD.
- Use the **Tab** key to auto-complete file names.
- Use **Up Arrow** to repeat previous commands.

9. Summary

This guide introduced basic computer operation, Windows usage, and essential CMD commands to help new students build confidence. Practice each section to improve your understanding.

10. Windows Interface (Step-by-Step Guide)

10.1 Windows Desktop

When you log into Windows, the first screen you see is the **desktop**. - **Icons**: Shortcuts to files, folders, and applications. - **Recycle Bin**: Stores deleted items until emptied.

10.2 Taskbar

Located at the bottom of the screen.

- **Start Menu Button**: Opens all apps and settings.
- **Search Bar**: Quickly find files and programs.
- **Pinned Apps**: Commonly used programs.

- **System Tray:** Shows time, network, sound settings.

10.3 Start Menu

- Access installed apps.
- Open **Settings**, **Power Options**, **Documents**, etc.

10.4 File Explorer (Step-by-Step)

To open File Explorer:

1. Click the folder icon on the taskbar 2. OR press **Windows + E**

Inside File Explorer: - **Quick Access:** Frequently used folders. - **This PC:** Shows drives (C:, D:).
- **Address Bar:** Shows your current location. - **Navigation Pane:** Allows moving between folders.

10.5 Creating Folders & Files (Graphical Method)

- Right-click → **New** → Folder
- Right-click → **New** → Text Document

11. Practical Labs / Weekly Activities

Lab 1: Windows Basics

- Turn on the computer and log in.
- Identify desktop components.
- Open File Explorer and navigate between folders.

Lab 2: Creating and Managing Files

- Create a folder named **MyWork**.
- Inside it, create **Task1.txt**.
- Open the file and write your name.
- Copy the file to another folder.

Lab 3: Basic CMD Operations

- Open Command Prompt.
- Move between drives.
- Create a folder using CMD.
- Display directory content.
- Delete a folder.

Lab 4: File Operations via CMD

- Create two files using copy con.
- Rename one file.

- Copy a file into a new directory.
- Use type to display file content.

Lab 5: Xcopy Practice

- Create a folder structure.
- Use xcopy to copy all folders and subfolders.

12. Exercises and Questions

12.1 Multiple Choice Questions

1. Which command is used to clear the CMD screen?
A) clr
B) clean
C) cls
D) clear
2. What does the command cd .. do?
A) Go to the root directory
B) Go up one level
C) Create a directory
D) Delete a directory
3. Which one is used to display the contents of a file?
A) show
B) type
C) list
D) read
4. What is the shortcut to open File Explorer?
A) Ctrl + E
B) Windows + F
C) Windows + E
D) Alt + E

12.2 True or False

1. rd can remove a directory even if it contains files.
2. The Start Menu contains installed programs and power options.
3. copy con is used to display file contents.
4. The desktop is the first screen you see after logging in.

12.3 Short Questions

1. Explain the purpose of the Taskbar.
2. Describe the difference between `dir /p` and `dir /w`.
3. Write the command to rename `test.txt` to `info.txt`.